CHAPTER 5

INFORMATION PROCESSING

9S4JQS

Learning Objectives

- To know how to represent numerical and algebraic expressions by tree diagrams.
- To know how to write numerical and algebraic expressions from tree diagrams.

5.1 Introduction

In today's digital era, it is almost impossible to imagine a day without computers. Right from small shops to big software companies, the use of computers is inevitable. If there are no computers, most of the works will be stopped. Computers are able to find solutions even for complicated numerical expression and algebraic expression in quick and easy way. The answer given by the computer will be very precise and need not to be recalculated. There will be a question, how the computer read these expression?

Yes, Computers use **Tree diagram** to perform billions of operations in a uniform way and gives the answer. In this chapter we will learn about the **Tree diagram for both numeric and algebraic expressions**.

MATHEMATICS ALIVE – INFORMATION PROCESSING IN REAL LIFE



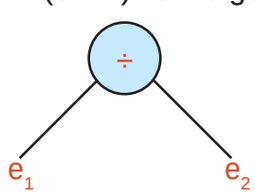
Information process in the Human Brain



Information process in computer

Consider the numerical expression $[(9-4)\times 8] \div [(8+2)\times 3]$. We can try to understand the expression in a better way through the tree diagram.

1) Let us consider $e_1 = (9-4)\times 8$, $e_2 = (8+2)\times 3$ we get



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